Dual Negative Selection for Anomaly Detection at Cold Start: A Normal-Only Comparison with PyOD

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Abstract

A newly installed sensor system may need an initial anomaly detector before it has accumulated a long operating history or observed a failure. We examine Dual Negative Selection Algorithm (Dual NSA) as a candidate for this cold start, using a small set of accepted normal observations and no real outliers during training. The evidence comes from 17 ODDS tabular datasets, five splits, two feature representations, and six training sizes, with 13 NSA configurations and 43 other PyOD entries. Four central tables compare Dual NSA on every dataset at 100%, 25%, 10%, and 5% of a capped normal training pool. In PCA(2), its mean AUROC is 0.7172, 0.7603, 0.7626, and 0.7264, respectively; its rank against 25 other detectors and itself improves from 13/26 to 10/26, 6/26, and 5/26. At 5%, models learn from 4–30 normals. A neural batch-size sensitivity expands the comparison field and leaves Dual NSA 5/27. These findings support Dual NSA as a competitive starting option under the tested constraints. KDE remains ahead in mean AUROC, average precision declines at the smallest sizes, and dataset-level results vary substantially. Newly commissioned sensors are a proposed application; the experiment does not evaluate chronological sensor streams or deployed alarm performance.

1 Introduction: detecting anomalies before a long history exists

Consider a sensor set that has just begun operation. An operator may be able to approve a short commissioning period as normal, but there may be no examples of equipment faults, unusual environmental conditions, or attacks. The system still needs a first way to flag readings that depart from that initial experience. We call this condition a *normal-only cold start*: a small accepted normal sample is available, while real anomalous training examples are absent.

Negative selection offers a direct way to formulate that task. It learns a representation of self from normal examples and constructs detectors for departures from that self [1]. The Dual NSA implementation evaluated here combines negative detectors with an enclosing model of normal clusters. This gives it two ways to score an unfamiliar observation: matching a negative detector and falling outside the learned normal envelopes. Both can be learned without examples of real failures. Many conventional anomaly detectors also learn from normal-only data, so the practical argument requires a comparison with those alternatives.

Our question is whether Dual NSA remains a useful candidate when very few accepted normal observations are available. We compare it with PyOD detectors [2] under identical normal-only preprocessing and fitting rules. We bring the dataset-level comparisons into the main paper because an aggregate rank alone cannot guide a choice for a new system. The four settings are the full capped normal pool as a reference and 25%, 10%, and 5% retention as increasingly restricted starting samples. Results at 75% and 50%, every other NSA configuration, and the full detector roster remain in the appendix.

The central observation is that Dual NSA becomes more competitive in the two-dimensional PCA

pipeline as normal training availability shrinks. Its AUROC rank improves in a fixed comparison field, and it produces qualified results in every PCA(2) test case. This gives a reason to include it among the initial methods considered for a new sensor set with trusted commissioning data. The case is conditional: KDE is stronger on average, some datasets favor other methods, and a better rank at 5% does not imply better absolute detection than at 10%.

We make three contributions. First, we provide four direct, five-split comparisons on all 17 datasets, with actual normal training counts and explicit comparison-field ranks. Second, we explain what these results support for a normal-only cold start and what remains untested for newly commissioned sensors. Third, we retain the broader 57,120-record experiment, 740 supplemental neural attempts, failures, and audits in an extensive appendix and a reproducible code-and-results repository. Dual NSA was chosen for this focused presentation after inspecting the benchmark; the analysis is exploratory.

2 Dual NSA and the cold start setting

2.1 The model learned from accepted normal observations

The tested NSA implementation belongs to the pinned PyOD contribution described in the appendix [3]. For a point x in the estimator's fitted coordinate system, negative hyperspheres with centers c_j and radii r_j give a signed matching score

$$s_{\text{neg}}(x) = \max_j \{r_j - \|x - c_j\|\}. \tag{1}$$

A positive score indicates membership in at least one negative detector. Dual NSA additionally constructs enclosing self-cluster balls with centers a_k and expanded radii R_k. Its envelope and final scores are

$$s_{\text{env}}(x) = \min_k \{\|x - a_k\| - R_k\}, \qquad s_{\text{dual}}(x) = \max\{s_{\text{neg}}(x), s_{\text{env}}(x)\}. \tag{2}$$

An observation outside every self ball has a positive envelope margin, even when the sampled negative spheres leave a region uncovered. This makes Dual NSA an understandable initial scoring rule: departures from the normal envelopes can be identified without waiting to collect labeled faults. It also makes an assumption that needs checking in the application. A legitimate but previously unseen operating state can lie outside a small learned envelope.

The benchmark evaluates the outcome of this geometry. It does not isolate the envelope term experimentally, measure detector-space coverage, or establish why a particular retained sample performs well. Any explanation linking the rank improvement to self-envelope geometry is a hypothesis for further study.

2.2 What cold start means in this experiment

The training sample is small and clean by construction. Labels are used to remove anomalous rows from the training partition; no anomaly labels are passed to the initial fitting call. This models an operator having accepted a short normal commissioning sample. It does not solve the problem of deciding which initial readings are trustworthy when the system is already faulty.

The proposed sensor use case is a set of newly commissioned sensors, for example several readings describing the same equipment or environmental condition. An initial normal-only detector can supply scores while more accepted operating data are collected. Our data are heterogeneous tabular benchmarks, not a chronological record of such a commissioning process. The sensor interpretation is therefore a motivated application of the tested constraint, rather than a validated deployment result. Sequential dependence, drift, operating-mode changes, and alarm thresholds require their own evaluation.

3 Experimental comparison

3.1 Shared training and test data

We use 17 public ODDS datasets [4]: Arrhythmia, Cardio, Glass, Ionosphere, Letter, Lympho, MNIST, Musk, Optdigits, Pendigits, Pima, Satellite, Satimage-2, Shuttle, Vertebral, Vowels, and WBC. Each of five seeds, 42–46, makes a stratified 60/40 train/test split. All training anomalies are discarded before any preprocessing or model fitting. The normal training pool is capped at 600 observations, and the separate mixed test set is capped at 2,000 while preserving its class mix.

Nested seed-fixed subsets retain 100%, 75%, 50%, 25%, 10%, or 5% of that capped normal pool. Parent row order is preserved, and test rows stay fixed as training availability changes. The four main tables show 100%, 25%, 10%, and 5%. Here 100% means the full *capped normal pool*, not all observations in the original dataset. At 10%, actual training samples contain 8–60 normals; at 5%, they contain 4–30. Every detector receives the same features for a dataset, seed, representation, and retained size. Initial `fit` calls receive normal features without a label array.

We evaluate both original features and PCA(2). Standardization and PCA are fitted anew on retained normals at each size; the held-out test rows use those fitted transformations. Thus the study compares the whole normal-only pipeline as data availability changes. PCA directions are not held fixed. The main dataset tables use PCA(2), where Dual NSA has complete coverage and its strongest comparative standing. Original-feature results are reported in the main discussion and in companion appendix tables.

3.2 Methods displayed and methods attempted

The full grid has 11 NSA strategy families represented by 13 configurations and 43 other registered tabular PyOD entries: $17 \times 5 \times 2 \times 6 \times 56 = 57{,}120$ primary records. Constructors, epochs, ensemble sizes, and the CPU budget are documented in the appendix. This is a comparison of the recorded implementations and settings; it does not describe optimally tuned versions of every detector.

The four main dataset tables display Dual NSA, KDE, KNN, Isolation Forest (IForest), and One-Class SVM (OCSVM). This fixed comparator set covers density estimation, distances, isolation, and a conventional one-class boundary. KDE and KNN also give demanding empirical references near the small-sample settings. These display choices were made after inspection and are descriptive. All four conventional methods have 85/85 qualified outcomes at every size. The rank column includes the wider fixed comparison field, so hiding other columns does not improve Dual NSA's reported position. Full comparisons and AP companion tables are provided in the appendix.

XGBOD and DevNet require labeled anomalies for their intended supervised training in this roster and are unsuitable for the specified input. That restriction does not apply to most PyOD methods. Other failures reflect dimensions, sample-size settings, construction errors, or the computation budget. We mark failed cases *unsuitable under the tested configuration*; the label describes whether a configuration worked, not its detection accuracy or whether a different constructor could work.

3.3 Scores, complete fields, and uncertainty

We report AUROC and average precision (AP); larger values are better. Dataset rows average five qualified split scores. Overall means weight the 17 dataset means equally. A primary score is qualified when the recorded fit plus scoring time is at most 15 seconds. The nominal Python timer sometimes failed to stop native callbacks, so this measured-time rule is a disclosed post hoc enforcement. Raw-success sensitivity results retain the valid over-budget scores. Failed runs remain unscored, and no missing result is replaced by zero.

The primary PCA(2) field contains the same 25 other detectors qualified in all 85 cases at every one of the six retained sizes, plus Dual NSA. Other NSA configurations are not included in this focal denominator. For dataset d, with five-split AUROC mean $A_{d,m}$, the reported Dual rank is

$$r_d = 1 + \sum_{m \in \mathcal{B}} \mathbf{1}\{A_{d,m} > A_{d,\text{dual}} + 10^{-12}\}, \qquad |\mathcal{B}| = 25. \tag{3}$$

Ranks use competition ties with tolerance 10^{-12}. A rank such as 5/26 is a position among 26 methods, not a score percentile or a rank against all 43 attempted PyOD entries. The rank fraction $100r/26$ expresses the position as a percentage of that fixed field. A separate neural training-batch sensitivity adds VAE to the complete PCA(2) field; those ranks use 27 methods and are labeled separately.

Paired differences match successful dataset/seed cells. Descriptive 95% intervals use 5,000 dataset-cluster bootstrap draws and are not reported for fewer than five matched datasets. The comparisons are exploratory and unadjusted for multiplicity. Twelve comparator implementations use query-batch-dependent scoring; the appendix identifies them and supplies an inductive sensitivity. No mixed test features enter initial fitting, but these inference semantics differ from Dual NSA's pointwise scoring.

4 Results: Dual NSA as an initial alternative

4.1 Aggregate standing and four dataset comparisons

Table 1 summarizes the same four settings examined in detail in Tables 2–5. The dataset tables keep every task in the comparison, including tasks where Dual NSA is weak. Bold scores identify a maximum only among the five displayed methods, whereas the Dual rank uses all 25 fixed conventional comparators.

The same four PyOD comparators appear in every dataset table: KDE (density), KNN (distance), IForest (isolation), and OCSVM (one-class boundary). All four were selected after inspecting benchmark results and each has complete six-size qualified coverage. The comparison is descriptive. Bold identifies the highest AUROC among the five displayed methods within that dataset, including ties. Dual's rank uses the full fixed field of 25 other detectors plus Dual (26 total), not just these four comparators. Scores require five measured-$\leq$ *15-second successes; a dash never denotes zero.*

Table 1: Equal-dataset macro summary for PCA(2). Dual and KDE entries pair AUROC/AP; KNN reports AUROC. Dual's AUROC rank and rank percentage appear together: primary field 26 methods, neural-batch sensitivity 27. Percentage is 100 times rank/field size, with lower values better. Each field is fixed across all six sizes; the sensitivity leaves Dual scores unchanged.

Retained	Fit normals	Dual A/P	KDE A/P	KNN AUC	Dual AUC rank	Adapted AUC rank
100%	84–600	0.7172/0.4718	0.7890/0.4817	0.7958	13/26 (50.00%)	13/27 (48.15%)
25%	21–150	0.7603/0.4699	0.7947/0.4711	0.8005	10/26 (38.46%)	10/27 (37.04%)
10%	8–60	0.7626/0.4206	0.7769/0.4373	0.7706	6/26 (23.08%)	6/27 (22.22%)
5%	4–30	0.7264/0.3755	0.7496/0.3949	0.7082	5/26 (19.23%)	5/27 (18.52%)

Table 2: PCA(2), 100 percent retained normals: five-seed mean AUROC on all 17 datasets; n counts retained training normals. Dual's rank uses the fixed 26-method field; rank percentage is 100 times rank/26 (lower is better). Total sums n for one split, reports dataset-macro scores, and ranks those means. Bold marks maxima among displayed methods.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual rank	Rank %
arrhythmia	231	0.6367	0.6594	**0.6659**	0.6543	0.6620	17/26	65.38%
cardio	600	**0.9424**	0.9376	0.9276	0.9343	0.9217	1/26	3.85%
glass	123	0.7622	0.7976	**0.8762**	0.7902	0.8488	13/26	50.00%
ionosphere	135	0.6933	0.7637	**0.7723**	0.6778	0.7713	11/26	42.31%
letter	600	0.3732	0.5900	**0.6455**	0.4976	0.5769	20/26	76.92%
lympho	84	0.8500	**0.8879**	0.8397	**0.8879**	0.8190	13/26	50.00%
mnist	600	0.6212	**0.7235**	0.6651	0.6924	0.6447	19/26	73.08%
musk	600	0.9995	**0.9999**	0.9998	0.9814	**0.9999**	7/26	26.92%
optdigits	600	0.3492	0.5304	**0.6062**	0.4323	0.5502	23/26	88.46%
pendigits	600	0.9102	0.9812	0.9745	0.9645	**0.9847**	14/26	53.85%
pima	299	0.6350	**0.7104**	0.6930	0.6935	0.6645	15/26	57.69%
satellite	600	0.8165	**0.8431**	0.8339	0.7955	0.8262	8/26	30.77%
satimage-2	600	0.9266	0.9823	**0.9847**	0.9682	0.9775	16/26	61.54%
shuttle	600	0.9787	0.9791	**0.9889**	0.9731	0.9819	10/26	38.46%
vertebral	126	**0.3819**	0.3583	0.3079	0.3599	0.3740	8/26	30.77%
vowels	600	0.4153	0.7244	**0.8196**	0.6759	0.7175	21/26	80.77%
wbc	213	0.9002	**0.9432**	0.9283	0.9291	0.9405	17/26	65.38%
Total	7211	0.7172	0.7890	**0.7958**	0.7593	0.7801	13/26	50.00%

Table 3: PCA(2), 25 percent retained normals: five-seed mean AUROC on all 17 datasets; n counts retained training normals. Dual's rank uses the fixed 26-method field; rank percentage is 100 times rank/26 (lower is better). Total sums n for one split, reports dataset-macro scores, and ranks those means. Bold marks maxima among displayed methods.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual rank	Rank %
arrhythmia	57	**0.7113**	0.6726	0.6888	0.6877	0.6652	1/26	3.85%
cardio	150	0.9274	**0.9295**	0.9179	0.9240	0.9249	4/26	15.38%
glass	30	**0.9122**	0.8726	0.8305	0.7976	0.9012	1/26	3.85%
ionosphere	33	0.7433	**0.7529**	0.7233	0.6488	0.7482	5/26	19.23%
letter	150	0.5155	0.5991	**0.6263**	0.5009	0.5773	15/26	57.69%
lympho	21	0.7310	0.8328	**0.8879**	0.8569	0.6879	16/26	61.54%
mnist	150	0.6448	**0.7420**	0.6968	0.6908	0.6774	18/26	69.23%
musk	150	0.9937	0.9984	**0.9990**	0.9664	0.9972	9/26	34.62%
optdigits	150	0.2992	0.6090	**0.6610**	0.4738	0.6272	24/26	92.31%
pendigits	150	0.9045	**0.9676**	0.9665	0.9305	0.9653	13/26	50.00%
pima	74	0.6606	**0.7190**	0.7012	0.6994	0.6721	17/26	65.38%
satellite	150	0.8179	**0.8339**	0.8251	0.7769	0.8107	6/26	23.08%
satimage-2	150	0.9664	**0.9767**	0.9744	0.9460	0.9712	9/26	34.62%
shuttle	150	**0.9834**	0.9757	0.9743	0.9575	0.9766	2/26	7.69%
vertebral	31	**0.4327**	0.3917	0.4077	0.3915	0.3847	9/26	34.62%
vowels	150	0.7654	0.7129	**0.8005**	0.6333	0.7328	4/26	15.38%
wbc	53	0.9153	0.9231	**0.9273**	0.9022	0.9196	10/26	38.46%
Total	1799	0.7603	0.7947	**0.8005**	0.7520	0.7788	10/26	38.46%

Table 4: PCA(2), 10 percent retained normals: five-seed mean AUROC on all 17 datasets; n counts retained training normals. Dual's rank uses the fixed 26-method field; rank percentage is 100 times rank/26 (lower is better). Total sums n for one split, reports dataset-macro scores, and ranks those means. Bold marks maxima among displayed methods.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual rank	Rank %
arrhythmia	23	0.6401	0.6397	**0.7097**	0.6841	0.6310	16/26	61.54%
cardio	60	0.8860	**0.9248**	0.9182	0.9218	0.9043	16/26	61.54%
glass	12	0.8470	0.8616	0.7512	0.7530	**0.8829**	5/26	19.23%
ionosphere	13	0.7478	**0.7515**	0.6548	0.6569	0.7511	5/26	19.23%
letter	60	0.6229	0.6275	**0.6398**	0.4944	0.6352	6/26	23.08%
lympho	8	0.7207	**0.7931**	0.7828	0.7862	0.7466	12/26	46.15%
mnist	60	0.6640	**0.6934**	0.6699	0.6927	0.6556	15/26	57.69%
musk	60	0.9797	**0.9860**	0.9808	0.8723	0.9831	8/26	30.77%
optdigits	60	0.5067	**0.5491**	0.5395	0.4234	0.5373	11/26	42.31%
pendigits	60	0.8600	**0.9439**	0.9173	0.8959	0.9326	15/26	57.69%
pima	29	0.6628	**0.7188**	0.7104	0.6902	0.6656	12/26	46.15%
satellite	60	0.8136	**0.8175**	0.8052	0.7256	0.8066	3/26	11.54%
satimage-2	60	**0.9721**	0.9685	0.9652	0.9034	0.9616	2/26	7.69%
shuttle	60	0.9660	0.9746	0.9677	0.9456	**0.9756**	10/26	38.46%
vertebral	12	**0.4613**	0.4105	0.4401	0.3965	0.3968	8/26	30.77%
vowels	60	**0.7337**	0.6559	0.7306	0.5305	0.6772	1/26	3.85%
wbc	21	0.8799	0.8910	**0.9170**	0.8602	0.8844	11/26	42.31%
Total	718	0.7626	**0.7769**	0.7706	0.7196	0.7663	6/26	23.08%

Table 5: PCA(2), 5 percent retained normals: five-seed mean AUROC on all 17 datasets; n counts retained training normals. Dual's rank uses the fixed 26-method field; rank percentage is 100 times rank/26 (lower is better). Total sums n for one split, reports dataset-macro scores, and ranks those means. Bold marks maxima among displayed methods.

Dataset	n	Dual	KDE	KNN	IForest	OCSVM	Dual rank	Rank %
arrhythmia	11	0.6220	0.6356	0.6546	**0.6567**	0.6337	14/26	53.85%
cardio	30	0.8666	**0.9086**	0.9064	0.9059	0.8971	16/26	61.54%
glass	6	0.7799	0.7835	0.3787	0.6604	**0.8192**	4/26	15.38%
ionosphere	6	0.7147	**0.7350**	0.4874	0.6197	0.7338	5/26	19.23%
letter	30	0.6014	0.5996	0.5797	0.4628	**0.6055**	2/26	7.69%
lympho	4	0.4121	0.5431	**0.7500**	0.6716	0.5362	24/26	92.31%
mnist	30	0.6626	0.7168	0.7264	**0.7387**	0.6595	18/26	69.23%
musk	30	0.9350	**0.9537**	0.9358	0.7970	0.9488	7/26	26.92%
optdigits	30	**0.6226**	0.5938	0.5308	0.4757	0.6070	1/26	3.85%
pendigits	30	0.8256	**0.8954**	0.8634	0.8842	0.8754	17/26	65.38%
pima	14	0.5903	0.6707	**0.6805**	0.6735	0.6240	16/26	61.54%
satellite	30	**0.8185**	0.8181	0.7859	0.6893	0.8097	1/26	3.85%
satimage-2	30	0.9544	0.9543	0.9408	0.8547	**0.9553**	4/26	15.38%
shuttle	30	**0.9616**	0.9575	0.9381	0.9242	0.9615	4/26	15.38%
vertebral	6	0.3853	**0.4407**	0.3081	0.3862	0.4282	19/26	73.08%
vowels	30	**0.7107**	0.6310	0.6538	0.4133	0.6633	1/26	3.85%
wbc	10	0.8856	0.9050	**0.9196**	0.8655	0.8975	12/26	46.15%
Total	357	0.7264	**0.7496**	0.7082	0.6870	0.7445	5/26	19.23%

4.2 What changes when the normal sample is scarce?

Dual NSA's PCA(2) AUROC rises from 0.7172 at the full capped normal pool to 0.7603 at 25% and 0.7626 at 10%, then declines to 0.7264 at 5%. Its rank improves across these settings from 13/26 to 10/26, 6/26, and 5/26. The corresponding 10% and 5% positions are top 23.08% and 19.23% of this fixed field. These improvements give a concrete reason to consider Dual NSA during a normal-only cold start. They do not establish that reducing data always helps it.

KNN provides a useful reference for the changing gap. Its PCA(2) AUROC is 0.7958 at 100% and 0.7706 at 10%; Dual NSA's mean deficit narrows from 0.0786 to 0.0080. The matched change in that relative gap is +0.0707, with descriptive interval [+0.013, +0.136]. KDE remains the strongest mean-AUROC method in the fixed field at both smallest sizes, scoring 0.7769 at 10% and 0.7496 at 5%. Dual-minus-KDE differences are −0.0143 [−0.032, +0.004] and −0.0232 [−0.045, −0.002]. Dual NSA is competitive here, while a density baseline remains a strong starting candidate.

The AP comparison places an important limit on the argument. Dual NSA's PCA(2) AP is 0.4718, 0.4699, 0.4206, and 0.3755 across the four displayed sizes. Relative to 100%, paired AP changes at 10% and 5% are −0.0511 [−0.0990, −0.0084] and −0.0962 [−0.1756, −0.0310]. Paired AUROC changes are +0.0454 [−0.0013, +0.1002] and +0.0092 [−0.0687, +0.0828]. Better ordinal standing at 5% coexists with lower AP and with lower AUROC than at 10%. A system that depends on a small queue of high-priority alerts needs to check precision and its operating threshold separately.

4.3 Dataset variation and the original-feature check

The four dataset tables show why the starting choice must be validated locally. At 10%, Dual NSA's fixed-field rank ranges from 1/26 on Vowels to 16/26 on Arrhythmia. At 5%, it ranges from 1/26 on Optdigits to 24/26 on Lympho. Optdigits has only 0.6226 AUROC despite its first-place relative position; Lympho has 0.4121. Comparative rank and absolute discrimination must both be considered. The results do not support a uniform top-rank claim across datasets.

With original features, Dual NSA achieves 85/85 qualified outcomes at both 10% and 5%. Its AUROC/AP pairs are 0.7792/0.5112 and 0.7900/0.5066, and its AUROC ranks against the fixed 19-other-detector field are 14/20 and 10/20. KDE is ahead, with AUROC 0.8781 and 0.8577. Dual NSA has only 81/85 qualified cases at 100% and 83/85 at 25%, so it has no complete four-setting native rank comparison. On the 81 cases matched to 100%, its AUROC changes at 10% and 5% are +0.0780 [+0.0339, +0.1305] and +0.0903 [+0.0410, +0.1436]. Companion appendix tables retain the incomplete cases and their counts. The especially favorable PCA(2) ranks should not be generalized to arbitrary sensor feature dimensions.

4.4 Does a small-sample repair change the comparison?

The pinned AutoEncoder and VAE use training batches of 32 with incomplete batches discarded. At 5%, every sample is smaller than 32, so the original loops receive no training batch. This is a sample-size implementation failure; it does not establish a need for training outliers. A separate 740-attempt sensitivity uses training batch size $\min(32, n_{\text{fit}})$ while preserving inference batches, splits, preprocessing, epochs, the scoring procedure, and budget. Both methods then qualify in all 85 cases of every new 10% and 5% panel.

The repair adds VAE to the complete six-size PCA(2) comparison field. Dual NSA's four AUROC ranks become 13/27, 10/27, 6/27, and 5/27, corresponding to top 22.22% at 10% and 18.52% at 5%. Its small-sample standing persists when that concrete comparator failure is repaired. A separate primary sensitivity removes query-batch-dependent scorers: the fixed inductive field contains 20 other detectors plus Dual NSA, giving ranks 13/21, 10/21, 6/21, and 5/21. This keeps the same ordinal positions while changing the rank fractions to top 28.57% and 23.81% at the two smallest sizes. The native fixed field does not expand after the neural repair because unrelated historical failures remain. Full outcomes, provenance, and a brief startup timing overlap are disclosed in the appendix.

4.5 How the other NSA mechanisms compare

Table 6 places the other tested NSA configurations beside Dual NSA using the same fixed PCA(2) field. Each NSA is inserted separately into the field, so its rank is directly interpretable against conventional alternatives. Dual NSA leads the complete NSA group at 10% and 5% by both mean AUROC and AP. The table also preserves the three binary configurations that fail to generate valid negative detectors in the tested PCA(2) construction.

All 13 NSA configurations are retained here. Each AUROC rank inserts the focal NSA into the same 25 other PyOD detectors with complete qualified coverage at all six sizes. The 5 percent scores are shown only for configurations with all 85 qualified dataset/seed outcomes; Q preserves every configuration's coverage. All three binary configurations are unsuitable under the tested PCA(2) construction, with 0/85 qualified outcomes. A missing rank or score is not zero.

Table 6: All 13 NSA configurations in PCA(2): AUROC rank against the same fixed 25 other PyOD detectors plus the focal NSA at each retained size. Five-percent AUROC/AP are complete-panel macro means; Q is five-percent qualified coverage out of 85.

NSA configuration	100% rank	25% rank	10% rank	5% rank	5% AUROC	5% AP	5% Q
NSA fixed	22/26	19/26	19/26	19/26	0.5917	0.2200	85/85
NSA variable	15/26	11/26	8/26	10/26	0.6683	0.2879	85/85
NSA coverage	15/26	14/26	9/26	10/26	0.6669	0.2838	85/85
NSA grid	15/26	11/26	8/26	10/26	0.6775	0.3002	85/85
NSA hierarchical	15/26	14/26	12/26	12/26	0.6579	0.2647	85/85
NSA voronoi	15/26	11/26	8/26	8/26	0.6882	0.3153	85/85
NSA deterministic	21/26	17/26	17/26	17/26	0.6066	0.2198	85/85
NSA suppressed	15/26	12/26	9/26	12/26	0.6650	0.2955	85/85
NSA dual	13/26	10/26	6/26	5/26	0.7264	0.3755	85/85
NSA annealed	21/26	19/26	17/26	19/26	0.5966	0.2187	85/85
NSA binary Hamming	—	—	—	—	—	—	0/85
NSA binary contiguous	—	—	—	—	—	—	0/85
NSA binary chunk	—	—	—	—	—	—	0/85

All ten continuous PCA(2) NSA configurations improve their relative AUROC ranks at 10% and 5% compared with 100%. At 10%, all ten increase their absolute AUROC point estimates; at 5%, only three do, and all ten have lower AP. The native-feature family results are much less favorable. The broader family evidence therefore supports a focused argument for Dual NSA under these constraints, with representation and strategy choice remaining central. The appendix provides every strategy, all six sizes, original-feature results, failure reasons, paired contrasts, and alternative rank fields.

5 Discussion: a starting detector for newly commissioned sensors

5.1 When the proposed use case fits

A sensor set that has just begun collecting data can have enough trusted readings to describe an initial normal state while having no real failure examples. Under that condition, Dual NSA can be a good alternative to include in the first detector comparison. The rationale combines its normal-only training rule, pointwise scoring, complete small-sample outcomes, and competitive standing in the tested PCA(2) pipeline. It can supply an initial anomaly score without making collection of labeled faults a prerequisite.

The practical role is an initial candidate whose performance is checked as data become available. KDE and KNN should remain demanding reference methods: neither requires real training outliers in this experiment, and both are strong comparators. The tables give an operator a basis for choosing candidates, rather than a reason to select the same detector for every newly installed system. A small trusted commissioning set must also cover the normal modes that matter. Otherwise, a new but legitimate mode may be flagged simply because it was absent from the starting sample.

For multichannel sensors, the favorable PCA(2) result suggests evaluating a compact normal-only representation when such a representation preserves meaningful operating variation. Dimensional reduction is part of the tested pipeline; we do not show that it is appropriate for every sensor set. The original-feature results provide a useful check against treating the PCA advantage as a property of all

NSA deployments. A single scalar sensor, a strongly time-dependent process, and a high-dimensional multichannel system may need different representations and comparison methods.

5.2 What the experiment establishes

The evidence establishes that the evaluated Dual NSA implementation works with very small normal-only training matrices and can rank near the front of a fixed field in PCA(2). It also establishes that this standing survives a targeted neural batch-size repair. The observation concerns comparative behavior under scarcity. At 5%, the improved rank partly reflects deterioration among competing methods, while Dual NSA itself loses performance relative to 10%.

No experiment here chooses an operational alarm threshold, measures false alarms per hour, or tests a chronological stream after commissioning. AUROC and AP are held-out ranking measures. PyOD's contamination value of 0.1 calibrates internal score thresholds; it does not add training anomalies and does not validate a 10% deployed alarm rate. Thus the benchmark supports a candidate for initial monitoring, with sensor-specific threshold and stream validation still needed before interpreting scores as reliable alarms.

Query-batch-dependent comparators also need a separate interpretation for streaming use. The twelve audited implementations are COF, LOCI, SOS, SOD, ECOD, COPOD, RGraph, SUOD, LMDD, DIF, CD, and ROD. They may use other observations in the mixed query batch when producing scores, while Dual NSA scores pointwise from its fitted state. Their benchmark outcomes are retained, and an inductive sensitivity removes those entries from its field. This distinction concerns inference semantics and does not mean that the normal-only initial fitting rule was violated.

5.3 Limits and next validation

The clean starting data are selected with known benchmark labels. Real commissioning data can be contaminated or incomplete. The random stratified splits do not model temporal drift or the order in which a sensor would encounter operating modes. Training and test caps, fixed constructors, reduced epochs for some models, timing load, and the 15-second qualification limit further bound the comparison. Failures indicate unsuitability under the recorded configuration; the AutoEncoder/VAE repair demonstrates why that scope matters.

A direct sensor study should next use a chronological accepted-normal commissioning period, hold out later normal modes and faults, and compare Dual NSA with the same strong alternatives. It should report calibrated false-alarm rates, precision under the actual anomaly prevalence, detection delay, and sensitivity to contaminated commissioning data. Holding the representation fixed in an ablation and measuring envelope coverage would help distinguish sample-size effects from changes in PCA geometry. These are proposed follow-up evaluations, not outcomes of the present benchmark.

6 Reproducibility and conclusion

The public code-and-results repository provides the executable benchmark and machine-readable evidence.

The tested source is pinned to commit `0d6c901652f82`; its full commit and tree identifiers are recorded in the appendix. Dataset and split hashes identify the inputs. Independent audits reconstructed the retained normal subsets and checked zero fitted outliers. Raw aggregate records, coverage, sanitized logs, recovery evidence, and reporting arithmetic are retained. Per-observation decision vectors were not saved, so the reporting audit cannot independently recompute AUROC or AP from predictions. The

appendix contains the complete experimental report, including both smaller and intermediate training sizes and every attempted method.

Dual NSA is a defensible alternative to consider at a normal-only cold start, particularly in the tested low-dimensional pipeline. With only 4–30 training normals at 5%, it ranks 5/26 in the primary PCA(2) field and 5/27 after the neural batch-size check. That result supports the proposed use case of newly commissioned sensor sets with accepted normal data. Dataset variation, AP losses, native-feature results, and strong density and distance baselines keep the argument specific: Dual NSA can be a useful starting candidate, and the sensor application still needs local, chronological validation.

References

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